Why Pooled and Task-Balanced Perplexity Disagree

271M | Pooled: -2.01%

504M | Pooled: +1.18%

906M | Pooled: -2.78%

passage_retrieval_en
musique
narrativeqa
repobench-p
2wikimqa
hotpotqa
multi_news
qmsum
lcc
gov_report
triviaqa
multifieldqa_en
samsum
qasper
passage_count
trec

−0.015 −0.010 −0.005 0.000 0.005 0.010 0.015

Contribution to Δ log PPL

271M | Task-balanced: +16.76%

504M | Task-balanced: +3.86%

906M | Task-balanced: -17.37%

passage_retrieval_en
musique
narrativeqa
repobench-p
2wikimqa
hotpotqa
multi_news
qmsum
lcc
gov_report
triviaqa
multifieldqa_en
samsum
qasper
passage_count
trec

−0.075 −0.050 −0.025 0.000 0.025 0.050 0.075

Contribution to Δ log PPL

Blue = dropout helps; red = no dropout helps. Contributions sum in log-PPL space.